

TWIST: An efficient spatial-temporal transformer with temporal window and sparse attention for traffic forecasting

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Abstract—Accurate traffic prediction is crucial for Intelligent Transportation Systems (ITS). However, recent advancements in deep learning have encountered significant challenges in effectively capturing long-range dependencies and modeling complex inter-variable relationships, particularly under real-time processing constraints. These limitations primarily arise from the inherent trade-off in current model architectures: while some are optimized for short-term forecasting with constrained receptive fields, others prioritize long-term prediction accuracy at the cost of computational efficiency. Moreover, the dynamic and intricate interactions among nodes present significant scalability challenges, increasingly hindering efficient modeling as the road network grows. To address these limitations, we propose TWIST (temporal window and sparse spatial attention transformer), a novel architecture specifically designed for traffic forecasting. The proposed framework incorporates two key innovations: (1) a trend-aware window attention mechanism in the temporal dimension that effectively and efficiently captures multi-scale temporal dynamics, and (2) a sparse attention mechanism in the spatial dimension that optimizes computational efficiency by selectively focusing on significant spatial nodes. Extensive experiments on eight real-world datasets demonstrate that TWIST consistently outperforms state-of-the-art methods across various scenarios, including regular, long-term and large-scale traffic forecasting tasks, while maintaining competitive computational efficiency. Our code is publicly available at <https://github.com/STGTraffic/TWIST>.

Index Terms—Spatial-temporal data, Attention mechanism, Traffic forecasting, Deep learning

I. INTRODUCTION

The rapid development of urban transportation systems has resulted in a substantial increase in vehicle numbers, leading to challenges such as congestion, environmental pollution, and traffic accidents. In response, Intelligent Transportation Systems (ITS) have emerged as a critical solution to address these issues [1]. Accurate traffic prediction plays a key role in ITS. By analyzing traffic data, researchers can build models to forecast future traffic conditions, providing useful insights to decision-makers. However, due to the inherent complexity

of spatiotemporal correlations in traffic data, accurate traffic forecasting remains a significant challenge.

Previous studies employed statistical methods, such as HA (Historical Average) [2] and ARIMA (Auto-Regressive Integrated Moving Average) [3], to predict future traffic data. However, the inherent complexity and nonlinearity of traffic data prevent these approaches from effectively capturing traffic patterns.

Recently, deep learning methods have been widely applied for traffic forecasting. Fig. 1 exhibits the generic architecture of the spatial-temporal forecasting models. In the temporal dimension, Recurrent Neural Network (RNN) [4] and its variants, such as Long Short-Term Memory (LSTM) [5] and Gated Recurrent Unit (GRU) [6], have been widely adopted to model time series in the early stage. In the spatial dimension, early researchers used Convolutional Neural Network (CNN) [7] to extract spatial correlations. These approaches model the road network as a regular grid structure, where each grid vertex is treated as a node. CNNs employ shared convolutional kernels to aggregate features from the central node and its neighboring nodes through weighted summation. However, real-world road networks exhibit non-Euclidean structures and varying degrees of connectivity among sensor nodes, making CNN-based methods struggle to capture spatial dependencies effectively. To address this problem, spatial-temporal graph neural networks (STGNNs) are explored to capture spatial-temporal correlations [8]–[10]. In contrast to CNNs, GNNs can effectively capture the intricate relationships between nodes, making them more effective in traffic forecasting. In subsequent studies, researchers further integrate STGNNs and attention mechanisms to strengthen the model's performance [11]–[13]. Despite their success in exploring the spatio-temporal characteristics of traffic forecasting, several limitations persist in current approaches.

Firstly, in the temporal dimension, existing models are typically designed and optimized for short-term predictions. As a result, these models often demonstrate suboptimal performance or significant computational inefficiencies when extended to long-term forecasting scenarios. Many existing approaches rely on RNNs or 1-D CNNs, which exhibit inherent limitations in modeling long-term temporal correlations. RNNs, in particular, are prone to suffer from vanishing or exploding gradients, significantly restricting their capacity to capture long-range correlations in time series data [14]. CNNs-based methods perform forward computation by sliding a 1-D convolutional kernel across consecutive symbols.

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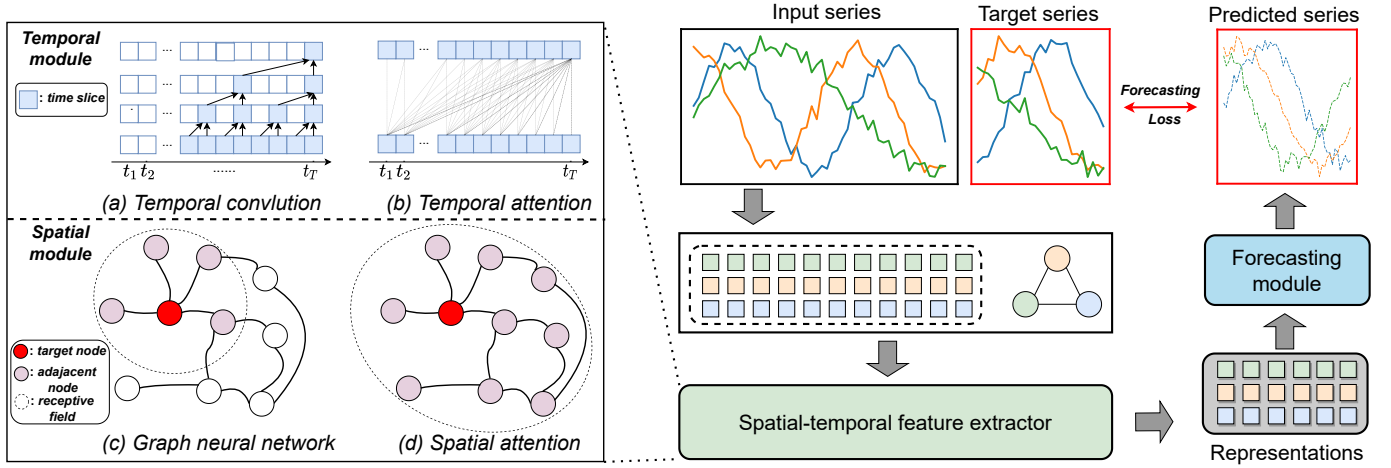


Fig. 1. A universal architecture of a spatial-temporal forecasting model. The input series is transformed into vector representations and then it is processed through the spatial-temporal feature extractor. The resulting spatial-temporal features are fed into the forecasting module to generate the predicted series. The forecasting loss is computed by comparing the predicted series with the target series. Temporal modules (a) and (b): TCN requires stacking multiple layers to capture long-term temporal dependencies, while standard temporal attention incurs $O(T^2)$ complexity for feature extraction. Spatial modules (c) and (d): GNN focuses on local spatial dependencies, and the standard spatial attention with $O(N^2)$ complexity can capture global dependencies but it is inefficient for large-scale datasets.

However, the limited localized convolutional kernels result in suboptimal performance when applied to long-term forecasting. To address this problem, some researchers employ Temporal Convolutional Network (TCN) to capture long-term temporal dependencies [9], [10], [13]–[16], shown in Fig. 1 (a). Although TCN can expand the receptive field through dilated convolutions, its layer-by-layer stacking architecture may still suffer from information dilution, making it difficult to capture long-range dependencies effectively [17]. Furthermore, transitioning from short-term to long-term forecasting often requires stacking numerous layers to capture global temporal dependencies, resulting in significant computational inefficiencies. Recent advancements have attempted to address these limitations by integrating attention mechanisms into traffic prediction frameworks [13], [17]–[20], as shown in Fig. 1 (b). Attention mechanisms provide a flexible and effective solution by focusing on the most relevant time steps, enabling the capture of long-term dependencies without relying on deep layer stacking like CNNs. However, despite these advantages, the computational complexity increases significantly, primarily due to the quadratic growth in pairwise computations between time steps. Consequently, the performance or efficiency of existing methods often degrade substantially as the forecasting horizon expands.

Secondly in the spatial dimension, existing spatial-temporal neural networks exhibit limited scalability to model the spatial correlations. As shown in Fig. 1 (c), recent approaches commonly employ graph-based methods, such as Graph Convolutional Network (GCN) and Graph Attention Network (GAT), however, these methods face inherent limitations. Specifically, the feature propagation in GCN is typically localized [13], as a result, information propagation from distant nodes requires the stacking of multiple layers, further leading to issues such as over-smoothing. This hinders the model's ability to capture dynamic and semantic-based spatial relationships. To

address this challenge, recent studies have focused on designing data-driven graphs to better adapt to evolving network structures and incorporate semantic information [21], [22]. However, the performance of graph-based models heavily relies on the accuracy and rationality of the input graph, as inaccurate or suboptimal node connections can propagate erroneous or irrelevant information. On the other hand, data-driven methods typically integrate dynamic traffic signals with learnable embeddings to construct dynamic adjacency matrices via inner product operations. However, this process incurs a computational complexity of $O(N^2)$, which poses significant challenges for model training, especially when dealing with large-scale datasets. As shown in Fig. 1 (d), spatial attention mechanisms offer an alternative approach by dynamically capturing relationships between nodes through attention weights, enabling more effective extraction of spatial features [23], [24]. Despite their flexibility, the quadratic computational complexity $O(N^2)$ associated with pairwise node interactions significantly limits their scalability and practical applicability in large-scale traffic forecasting tasks.

Motivated by the aforementioned limitations, we propose *TWIST*, an efficient and effective end-to-end spatiotemporal model for traffic prediction. To address the first limitation, we introduce a novel trend-aware window attention mechanism. This module utilizes 1-D CNN to generate queries and keys, ensuring robust performance even in challenging long-term forecasting scenarios. Furthermore, we divide the temporal dimension into non-overlapping windows, enabling the model to efficiently capture local dependencies within each window. By stacking multiple attention layers with varying window sizes, this mechanism facilitates multi-scale temporal modeling, allowing the model to learn both short-term and long-term trends effectively. To address the second limitation, we propose an efficient sparse query mechanism that reduces computational complexity to $O(N \log N)$, maintaining a global

receptive field while improving the model's scalability. Compared to existing approaches, our model does not rely on pre-defined or data-driven graphs. Due to the optimized attention mechanisms, it achieves both high prediction accuracy and computational efficiency. Extensive experiments conducted on eight real-world datasets demonstrate that TWIST consistently outperforms state-of-the-art methods in terms of accuracy and efficiency, exhibiting its superiority in traffic forecasting tasks.

Our main contributions can be summarized as follows:

- We propose TWIST, a fully attention-based model designed to effectively capture spatiotemporal dependencies in traffic data. Unlike existing approaches, it completely relies on the optimized attention mechanism and maintains high computational efficiency, filling a gap in current research.
- We design a novel temporal window attention mechanism, that can fully utilize local context to generate queries and keys, with adjustable window sizes to capture multi-scale temporal dependencies, facilitating long-term forecasting.
- In the spatial dimension, a sparse query sampling strategy is proposed, which reduces computational complexity to $O(N \log N)$. By using sampled nodes as sparse queries to aggregate features from global nodes, enabling efficient extraction of key spatial relationships, especially in large-scale traffic scenarios.
- Extensive experiments on eight real-world datasets demonstrate that TWIST achieves state-of-the-art performance in regular, long-term and large-scale traffic forecasting tasks while maintaining superior computational efficiency.

The structure of this paper is as follows: Section II reviews related work on traffic prediction. Section III defines the problem. Section IV details the method of our model. Section V presents experiments conducted on eight real-world datasets and analyzes the results. Finally, Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

Table I provides a comprehensive summary of existing models in related work, highlighting their spatiotemporal feature extraction methodologies.

A. Traditional methods for traffic forecasting

Early research treated traffic series as a form of time series and applied traditional methods for modeling, such as historical average (HA), support vector regressive (SVR) [45], and Kalman filter (KF) [46], etc. However, these methods including ARIMA and KF, are often limited by their reliance on linear assumptions, rendering them inadequate for capturing the nonlinear dynamics inherent in traffic data. Furthermore, traditional approaches primarily focus on temporal patterns while neglecting the spatial dependencies inherent in traffic networks, leading to decreased prediction accuracy.

B. Spatial-temporal neural networks for traffic forecasting

With the advancement of data availability, hardware capabilities, and algorithmic innovations, deep learning methods have become increasingly powerful and efficient, enabling the processing of large-scale datasets and the development of sophisticated architectures. Ma et al. first utilized LSTM for traffic forecasting [47] while Duan et al. used it for travel time estimation [48]. Subsequent studies introduced hybrid models, such as ST-ResNet [49] and Conv-LSTM [5], which combined CNNs for spatial feature extraction with LSTMs for temporal modeling. However, CNN-based methods are inherently limited by their reliance on Euclidean structures, making them unsuitable for the non-Euclidean nature of traffic networks.

To address this limitation, graph-based approaches emerged, leveraging the natural graph structure of road networks. Li et al. proposed DCRNN which uses bidirectional random walks to capture spatial correlations [8]. Guo et al. proposed GC-LSTM, a sequence-to-sequence model that integrates spectral graph convolution with LSTM for traffic speed prediction [50]. Considering that RNN-based methods may suffer from gradient vanishing or explosion, researchers have developed temporal convolutional networks (TCNs) to capture temporal dependencies. Yu et al. combined TCN and GCN to extract spatial-temporal features [25]. Wu et al. proposed a self-learning matrix construction method based on node embeddings and combined GCN with dilated causal convolution for traffic prediction [9]. Zhao et al. introduced the T-GCN, which combined the GCN with GRU [28]. Despite these advancements, many methods rely on static or predefined graphs, limiting their ability to capture dynamic spatial dependencies.

In the subsequent studies, researchers started to model dynamic spatial correlations by constructing data-driven graph structures. Li et al. constructed the dynamic graph based on temporal similarity between nodes [15]. Han et al. assumed that a shared graph could represent traffic patterns at the same time of day [10]. They employed three learnable matrices along with a core tensor to capture time-specific spatial dependencies among road segments. These components enabled the model to aggregate hidden states from neighboring nodes to target nodes through message-passing on dynamic adjacency matrix. However, assuming a shared graph for the same time of day oversimplifies dynamic traffic patterns. The use of learnable matrices and tensors adds computational overhead, reducing efficiency on large-scale networks, while local message passing limits capturing long-range dependencies. Ta et al. proposed Ada-STNet that learned graphs from both micro and macro aspects, reflecting comprehensive spatial correlations [51]. Guo et al. computed the strength of dynamic spatial correlations between nodes and then adjusted the static weight matrix using an element-wise dot product operation to capture these dynamic relationships [17]. Although this approach can model dynamic spatial associations to some extent, it still relies on static adjacency matrices, and in addition, the dot product-based approach carries $O(N^2)$ complexity. Lan et al. constructed a data-driven dynamic spatial-temporal aware graph by calculating the spatial-temporal aware distance of the

TABLE I
SUMMARY OF EXISTING METHODS

Methods	Temporal module	Spatial module	Predefined graph
DCRNN [8]	GRU	Diffusion GCN	Yes
STGCN [25]	TCN	GCN	Yes
GraphWaveNet [9]	TCN	GCN	Yes
ASTGCN [26]	TCN + Attention	GCN	Yes
STMGCN [27]	RNN	Multigraph GCN	Yes
T-GCN [28]	GRU	GCN	Yes
AGCRN [29]	GRU	GCN	Yes
STSGCN [30]	GCN	GCN	Yes
STFGNN [15]	TCN + GCN	GCN	Yes
DMSTGCN [10]	TCN + GCN	GCN	No
STGODE [31]	TCN	GCN	Yes
DGCN [14]	TCN + LSTM + Attention	Dynamic GCN + Attention	No
Traffic Transformer [32]	Transformer	Transformer	Yes
ASTGNN [17]	Transformer	Transformer + GCN	Yes
DDGCRN [33]	GRU	GRU	No
TE-TCN [34]	GRU + TCN + Transformer	GCN	No
STAEFormer [35]	Attention	Attention	No
PDFormer [36]	Attention	Attention	Yes
DSTET [37]	Attention	Attention	Yes
SAGDFN [38]	GRU	Attention	No
ProSTformer [39]	Attention	Attention	No
TC-GCN [40]	Attention	Attention + GCN	No
LCDFormer [41]	Transformer	Transformer	Yes
PGCN [42]	TCN	GCN	Yes
STPGNN [16]	TCN	GCN + Pivotal GCN	No
HimNet [43]	GRU	GCN	No
STAMT [44]	Temporal Aggregation Conv	Transformer	No

TABLE II
COMPARISON OF THE COMPUTATIONAL COMPLEXITY OF RECENT SPATIOTEMPORAL TRANSFORMER. T: INPUT LENGTH; N: NUMBER OF NODES. K: KERNEL SIZE. W: WINDOW SIZE.

Model	Temporal Complexity	Spatial Complexity
Traffic Transformer	$O(T^2)$	$O(N^2)$
ASTGNN	$O(T^2)$	$O(N^2)$
STAEFormer	$O(T^2)$	$O(N^2)$
PDFormer	$O(T^2)$	$O(N^2)$
DSTET	$O(T^2)$	$O(N^2)$
LCDFormer	$O(T^2)$	$O(N^2)$
STAMT	$O(T \cdot K)$	$O(N)$
TWIST (ours)	$O(T \cdot W)$	$O(N \log N)$

traffic data [52]. Wang et al. proposed the TC-GCN to enhance traffic flow forecasting by modeling cross-dimensional dependencies among channel, time, and space domains [40]. Shin et al. introduced PGCN, utilizing parameterized cosine similarity between node signals to model progressive spatial correlations on graphs [42]. Kong et al. proposed STPGNN, integrating the pivotal graph and TCN to effectively capture complex spatio-temporal dependencies [16]. However, both methods incur quadratic computational complexity when constructing dynamic adjacency matrices.

Attention mechanisms, initially developed for machine translation, have also been adapted for traffic prediction due to their ability to adaptively focus on relevant inputs. Liu et al. introduced the STAEformer model, which leverages a novel adaptive embedding and spatial-temporal attention to effectively capture traffic patterns [35]. Jiang et al. considered time propagation delay and designed the attention-based spatial block to capture spatial dependencies [36]. Sun et

al. introduced a novel decoupled spatial-temporal embedding structure that effectively incorporates low-frequency features into the sequences [37]. Cai et al. employed dual graph masking spatial attention to model compound spatial correlations, and trend-aware multi-head attention for temporal feature extraction [41]. Yan et al. proposed ProSTformer, a progressive spacetime self-attention mechanism designed to enhance traffic flow forecasting by efficiently focusing on relevant spatial and temporal dependencies [39]. Liu et al. proposed STAMT, which integrates temporal aggregation convolution and spatial attention for spatiotemporal feature extraction [44]. Although temporal aggregation convolution improves computational efficiency by compressing temporal dimension information, it may result in the loss of complete temporal details, which are crucial for long-term predictions. Table II summarizes the computational complexity of recent spatial-temporal transformer models, and it is observed that most of these methods incur high computational costs, particularly for long-term or large-scale prediction.

Compared with the studies mentioned above, our proposed TWIST has the following advantages and distinctions: (1) we design a trend-aware temporal window mechanism that reduces computational complexity while enhancing long-term prediction performance, (2) we propose a novel sparse query sampling strategy that optimizes spatial complexity to $O(N \log N)$, ensuring scalability for large networks, (3) our TWIST achieves superior prediction performance without stacking multiple layers.

III. PROBLEM STATEMENT

A. Notations and definitions

Traffic Network: A road network can be represented as a graph $G = (V, E, A)$, where V is the set of nodes, with $|V| = N$ denoting the number of nodes, E is the set of edges, and $A \in \mathbb{R}^{N \times N}$ represents the connectivity between nodes.

Traffic Signals: The historical traffic series at time slice t is represented as a graph signal $X_t \in \mathbb{R}^{N \times F}$, where F denotes the dimension of each node's attributes, such as traffic speed, volume, or occupancy.

B. Problem definition

Traffic Forecasting: Given T historical graph signals $X = [X_{(t-T+1)}, \dots, X_{(T)}] \in \mathbb{R}^{T \times N \times F}$, our goal is to learn a function F to forecast T future graph signals $Y = [Y_{(t+1)}, \dots, Y_{(t+T)}] \in \mathbb{R}^{T \times N \times F}$ as accurately as possible. The forecasting process can be formulated as Eq. 1:

$$[X_{(t-T+1)}, \dots, X_{(T)}] \xrightarrow{F(\cdot)} [Y_{(t+1)}, \dots, Y_{(t+T)}] \quad (1)$$

IV. METHODOLOGY

In this section, we first present the architecture of the proposed TWIST, as illustrated in Fig. 2. Our TWIST consists of three main modules: temporal embedding layer (TE), temporal trend-aware window attention (TAWA), and sparse query attention (SQA).

The TE module enhances the perception of periodicity and trends in traffic data through adaptive learnable embeddings. The TAWA module captures complex and dynamic temporal dependencies in high-dimensional spatiotemporal representations. It utilizes a window mechanism to reduce computational complexity and enable multi-scale temporal feature extraction through stacked attention layers with varying window sizes. The SQA module optimizes spatial attention computation by selectively focusing on relevant spatial nodes, reducing complexity while preserving the model's ability to capture critical spatial dependencies. Detailed descriptions of each component are provided below.

A. Temporal embedding layer

Traffic data typically exhibits strong temporal dependencies. For example, traffic patterns during peak and non-peak hours differ significantly [35], [43], [44]. We employ the temporal embedding layer to enhance the influence of periodicity and trends on traffic data, shown in Fig. 2.

The embedding layer consists of two learnable embedding dicts, denoted as day-of-week embedding dict $E_w \in \mathbb{R}^{7 \times \varphi}$ and time-of-day dict $E_d \in \mathbb{R}^{t_d \times \varphi}$. In this work, 7 and t_d are the number of days in one week and the time steps in one day, respectively, and φ is the dimension of embedding. We denote $W_t, D_t \in \mathbb{R}^T$ as the day-of-week and time-of-day data for the traffic series from time step $(t - T + 1)$ to t . They are used as indices to obtain the day-of-week embedding $E_w^t \in \mathbb{R}^{T \times \varphi}$ and the time-of-day embedding $E_d^t \in \mathbb{R}^{T \times \varphi}$ from the embedding

dicts. Finally, we broadcast and concatenate them to get the temporal embedding by equation 2:

$$E_T = \text{Concat}(E_d^t, E_w^t) \quad (2)$$

where $E_T \in \mathbb{R}^{T \times N \times 2\varphi}$ is final embedding.

B. Trend-aware temporal window attention

In standard temporal self-attention, the computation of pairwise attention scores across all time steps results in quadratic complexity with respect to sequence length, significantly limiting its applicability in traffic prediction tasks. Furthermore, when the historical sequence length is extended, the relevance of distant data diminishes, introducing noise and uncertainty. It may cause the model to excessively focus on irrelevant or anomalous data, which can degrade performance.

To address these challenges, this work proposes a novel trend-aware window attention mechanism. Specifically, we divide the historical sequence into non-overlapping temporal windows and perform the attention computations within these windows. This operation can significantly reduce computational complexity while enabling efficient processing of long time series. Moreover, by stacking multiple attention layers with different window sizes, the model can obtain a multi-scale receptive field. As Fig. 3 shows, considering the limitations of point-wise dot product operations in standard attention, which often fail to capture local context. We employ 1-D CNN to generate queries and keys. This approach enhances the node's ability to recognize local trends and mitigate anomalies, facilitating long-term forecasting tasks.

Mathematically, given the original input $X \in \mathbb{R}^{T \times N \times F}$, we firstly obtain the high-dimension representation $H \in \mathbb{R}^{T \times N \times C}$ through linear transformation:

$$H = XW \quad (3)$$

where $W \in \mathbb{R}^{F \times C}$ is a learnable parameter matrix and C is the feature dimension. Then, to enhance the model's ability to capture local trends, we use 1-d CNNs along the temporal dimension to generate Q (query), K (key):

$$Q = \Theta_Q \star H, K = \Theta_K \star H \quad (4)$$

where Θ^Q and Θ^K are the parameters of convolution kernels with size k . Furthermore, we split the temporal dimension T into $t_n = T/T_w$ non-overlapping windows, each with a size T_w . For the i -th temporal window, we extract the corresponding slices of Q and K as:

$$\begin{aligned} Q_i &= Q[:, :, i \cdot T_w : (i+1) \cdot T_w], \\ K_i &= K[:, :, i \cdot T_w : (i+1) \cdot T_w] \end{aligned} \quad (5)$$

where $Q_i, K_i \in \mathbb{R}^{T_w \times N \times d_{model}}$ are the query and key in the i -th window. The attention computation for the i -th window is performed as:

$$\text{Attention}(Q_i, K_i, V_i) = \text{softmax} \left(\frac{Q_i K_i^T}{\sqrt{d_{model}}} \right) V_i \quad (6)$$

Here, $V_i \in \mathbb{R}^{T_w \times N \times d_{model}}$ is the value matrix derived by splitting $V = HW^V$, where $W^V \in \mathbb{R}^{C \times d_{model}}$. Finally, the

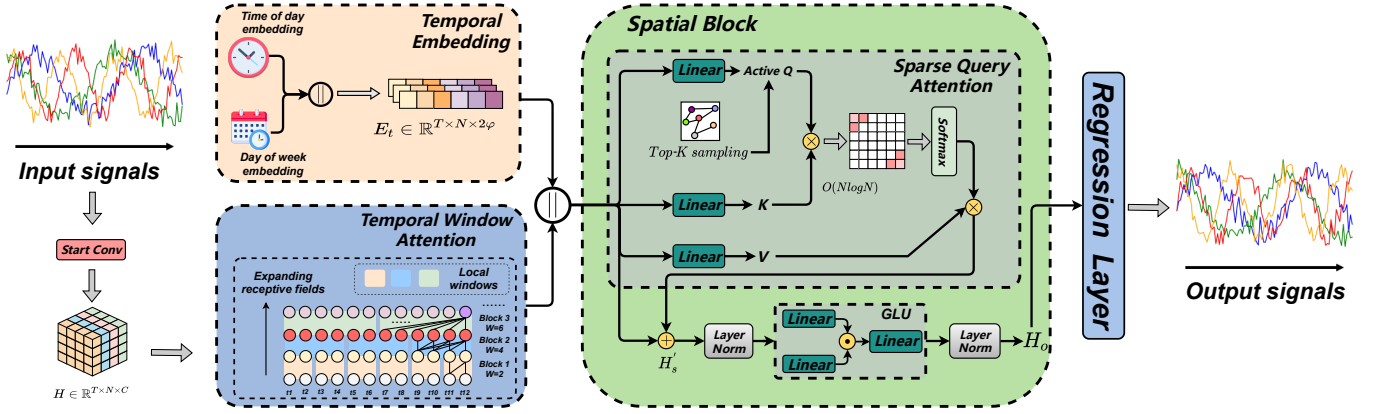


Fig. 2. The architecture of the proposed TWIST. GLU: gated linear unit. Linear: 1×1 convolutional layer for linear transformation.

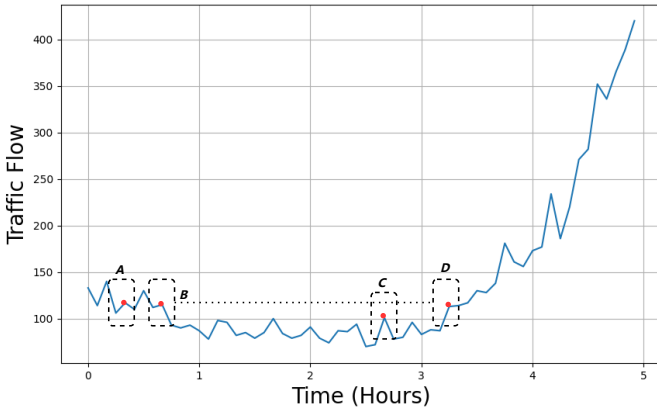


Fig. 3. In short-term forecasting, A and B tend to exhibit similar patterns as well as C and D. When the sequence is extended, although B and D have similar values, their trend directions are opposite. However, the standard attention mechanism assigns higher weights to these time steps with similar values, ignoring their differences in overall trends.

outputs WA_i from all windows are concatenated and projected back to the original feature space:

$$TAWA = \text{Concat}(WA_1, WA_2, \dots, WA_{t_n})W^T \quad (7)$$

where $W^T \in \mathbb{R}^{d_{\text{model}} \times C}$ is the learnable projection matrix.

We concatenate $TAWA$ and temporal embedding E_T to obtain updated representation $H_t \in \mathbb{R}^{T \times N \times (C+2\varphi)}$ for the downstream processing:

$$H_t = \text{Concat}(TAWA, E_T) \quad (8)$$

In this work, φ is set to $\frac{C}{2}$, and we denote $H_t \in \mathbb{R}^{T \times N \times 2C}$.

C. Sparse query attention

In traffic prediction, standard spatial attention mechanisms typically exhibit $O(N^2)$ computational complexity, leading to significant inefficiencies in model training and inference, especially on large-scale datasets. Similarly, data-driven graph construction often requires computing pairwise relationships among all nodes, further exacerbating computational costs. This issue becomes increasingly pronounced as the size of the dataset grows. In this work, we propose a novel query

sampling strategy to address this challenge. It can effectively reduce computational complexity while preserving a global receptive field. Specifically, query vectors are categorized into active and passive queries. Active query nodes actively sample and aggregate features across the network, capturing essential spatiotemporal patterns, while passive queries approximate their features based on the most correlated active query node, avoiding full attention computations. This approach is motivated by the hierarchical nature of traffic systems, where roads within the same region often exhibit similar patterns due to shared infrastructure and dynamics [53].

Given the H_t as input, we generate Q , K and V through linear transformation like standard spatial attention:

$$Q = W^Q H_t, K = W^K H_t, V = W^V H_t \quad (9)$$

where W^Q , W^K and $W^V \in \mathbb{R}^{2C \times d}$ are learnable parameter matrices. Then, we introduce the evaluation process for active queries and passive queries:

(1) **Key vectors (K) Sampling.** We random select the Keys and denoted selected Keys as K' , where $K' \in \mathbb{R}^{T \times N_{\text{sampler}} \times d}$ and d is the dimension of feature by linear mapping. The number of sampled nodes is calculated as $N_{\text{sampler}} = e \ln N$, where e is a predefined parameter and N is the total number of nodes.

(2) **Evaluation of queries.** We evaluate the significance of query vectors among all sampled key vectors by calculating the entropy [54], [55]. The entropy measures the uncertainty in the distribution of correlation scores between a query and sampled keys. A lower entropy indicates that the query has a more concentrated correlation with specific keys, making it more significant and active. Mathematically, let the i -th query vector be denoted as Q_i , where $Q_i \in \mathbb{R}^{T \times 1 \times d}$, and the j -th sampled key be $K'_j \in \mathbb{R}^{T \times 1 \times d}$. The correlation score matrix C_{ij} between query and sampled key is calculated by:

$$C_{ij} = Q_i (K'_j)^T \quad (10)$$

We obtain the probability distribution P_{ij} by using the exponential function over the score matrix:

$$P_{ij} = \frac{\exp(C_{ij})}{\sum_{K'_k \in K'} \exp(C_{ik})} \quad (11)$$

Based on the above equations, we define an activity function $H(C_{ij})$ to distinguish active and passive query vectors:

$$H = - \sum_{K'_j \in K'} \frac{\exp(C_{ij})}{\sum_{K'_k \in K'} \exp(C_{ik})} \log \left(\frac{\exp(C_{ij})}{\sum_{K'_k \in K'} \exp(C_{ik})} \right) \quad (12)$$

(3) **Top-U queries sampling.** We select the Top-U nodes from queries with the lowest entropy as active queries, where $U = N_{sampled}$:

$$\text{Active } Q = \text{Top}_U(-H(Q_i)) \quad (13)$$

For active queries: After obtaining the sampled active queries, we calculate the attention by:

$$\begin{aligned} SQA_{\text{active}} &= \text{Attention}(Q, K, V) \\ &= \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V \end{aligned} \quad (14)$$

For passive queries: In equation 14, $QK^T \in \mathbb{R}^{K \times N}$ represents the correlation matrix between sampled queries and all keys, and we denote it as matrix A . For each passive query (j) in the range $j \in \{U+1, U+2, \dots, N\}$, we find the most correlated active query node Q'_{idx_j} through matrix indexing:

$$idx_j = \arg \max_{i \in \{1, 2, \dots, U\}} A_{i,j} \quad (15)$$

where idx_j is the index of the active query Q'_{idx_j} that has the highest correlation with the passive query node j . $\arg \max$ is the function that finds the index of the maximum value in the j -th column in matrix A , corresponding to the most relevant active node for the j -th unsampled node. Finally, the output feature of j -th passive query is directly assigned from the attention output of its most correlated active query:

$$SQA_{\text{passive}_j} = SQA_{\text{active}_{idx_j}} \quad (16)$$

D. GLU and Prediction layer

In transformer networks, the gated linear unit (GLU) has been shown to be an effective replacement for the traditional Feed-Forward Network (FFN) due to its ability to introduce a gating mechanism, which enhances the model's feature selection capability [56]. Therefore, we use GLU to enhance the representation capability of proposed TWIST:

$$\begin{aligned} H_o &= GLU(H'_s)W_o \\ &= (\sigma(W_u H'_s) \odot (W_v H'_s))W_o \end{aligned} \quad (17)$$

where H'_s is the output of spatial attention, W_u , W_v , and $W_o \in \mathbb{R}^{2C \times 2C}$ are learnable parameter matrices for linear transformation. Finally, We employ the MLP to generate the prediction results:

$$Y = \text{MLP}(H_o) \quad (18)$$

where $Y \in \mathbb{R}^{T \times N \times F}$ represents the traffic signal to be predicted.

V. EXPERIMENTS

In this section, we conduct experimentation across eight real-world datasets to answer the following seven research inquiries.

- **RQ1:** How does TWIST perform compared to state-of-the-art models?
- **RQ2:** How efficient is TWIST compared to the baseline models?
- **RQ3:** What impact do the individual components have on the model's performance?
- **RQ4:** Whether the proposed method provides meaningful improvements over existing methods?
- **RQ5:** How robust is TWIST under different conditions?
- **RQ6:** What about the effect of the parameter on TWIST?
- **RQ7:** How does TWIST perform in real-world scenarios?

A. Datasets

To demonstrate the validity of the proposed method, we conduct experiments on eight real-world datasets: **PEMS03**, **PEMS04**, **PEMS07**, **PEMS08** [41], which represent coarse-grained traffic scenarios, as well as **NYCTaxi** and **NYCBike** [44], which represent fine-grained traffic scenarios. **Carpark** [38] and **GBA** [44] are large-scale traffic datasets. These datasets enable a comprehensive evaluation of our model's performance under various traffic conditions.

PEMS03, PEMS04, PEMS07, and PEMS08 are highway traffic datasets derived from the Performance Measurement System (PEMS) of the California Department of Transportation (CalTrans), including three traffic measures: traffic speed, flow, and occupancy. NYCBike and NYCTaxi are urban traffic demand datasets. NYCBike captures the bike-sharing patterns by recording the number of bike pickups and drop-offs at each station. NYCTaxi records taxi trip data across New York City, including details such as pickup and drop-off. These datasets provide valuable insights into urban mobility and traffic demand in a densely populated environment. The Carpark dataset consists of parking lot availability information for 1918 carparks in Singapore, with a time step of 5 minutes [38]. Unlike traditional traffic speed and flow datasets, Carprk focuses on parking lot availability, offering a unique perspective for evaluating prediction models in this domain. GBA is sourced from the PEMS of Caltrans, including traffic data from 2,352 sensors located across 11 counties in the Greater Bay Area. These two datasets provide large-scale traffic forecasting scenarios. More details are shown in Table III.

TABLE III
THE DETAILED STATISTICS OF EIGHT DATASETS

Dataset	Nodes	Interval	Samples	Time Range
PEMS03	358	5 min	26208	09/01/2018–11/30/2018
PEMS04	307	5 min	16992	01/01/2018–02/28/2018
PEMS07	883	5 min	28224	05/01/2017–08/31/2017
PEMS08	170	5 min	17856	07/01/2016–08/31/2016
NYCBike	250	30 min	4368	04/01/2016–06/30/2016
NYCTaxi	266	30 min	4368	04/01/2016–06/30/2016
Carpark	1918	5 min	17568	05/01/2016–06/30/2016
GBA	2352	15 min	35040	01/01/2019–12/31/2019

B. Metrics

For metrics, this work adopts three commonly used performance metrics, including Mean Absolute Error (**MAE**), Root Mean Squared Error (**RMSE**), and Mean Absolute Percentage Error (**MAPE**). MAE denotes the average of the absolute values of the predicted and true values. RMSE denotes the rooted average squared difference between the predicted values and the ground truth. MAPE is the mean absolute percentage error. The metrics are defined as follows:

- $MAE = \frac{1}{M} \sum_{i=1}^M |\hat{Y} - Y|$
- $RMSE = \sqrt{\frac{1}{M} \sum_{i=1}^M (\hat{Y} - Y)^2}$
- $MAPE = \frac{100}{M} \sum_{i=1}^M \left| \frac{\hat{Y} - Y}{Y} \right|$

where Y denotes the true values, $\hat{Y}$ denotes the predicted values, and M represents the number of values to be predicted.

C. Baseline models

To verify the validity of our method, we compare TWIST with the classical methods and state-of-the-art methods.

- **HA** [2]: HA is a classical time series forecasting model that uses historical average data to predict future traffic state.
- **VAR** [57]: VAR predicts future values by capturing linear interdependencies among multiple time series.
- **STGCN** [25]: STGCN is a model based on convolutional operations, integrating graph convolution with a 1-D temporal convolutional network to capture spatial-temporal dependencies.
- **DCRNN** [8]: DCRNN models the interactions between nodes as a diffusion process, introducing a diffusion convolution operation to capture spatial dependencies.
- **GraphWaveNet** [9]: GraphWaveNet captures spatial-temporal dependencies by combining adaptive graph structures with dilated causal convolution.
- **ASTGCN** [26]: ASTGCN leverages temporal attention and spatial convolution to capture dynamic spatial-temporal correlations.
- **AGCRN** [29]: AGCRN uses adaptive graph learning module to capture spatial dependencies in traffic networks and GRU is utilized to extract temporal features.
- **STG-NCDE** [58]: STG-NCDE combines neural controlled differential equations to model spatial-temporal dependencies in traffic forecasting effectively.
- **ASTGNN** [17]: ASTGNN introduces a trend-aware multi-head attention mechanism for traffic forecasting and develops a dynamic GCN to extract spatial correlations.
- **Ada-STNet** [51]: Ada-STNet employs a self-learning component to derive an optimal graph adjacency matrix from both macro and micro aspects.
- **DGCRN** [21]: DGCRN employs hyper-networks to extract dynamic characteristics from node attributes and generates dynamic graph for traffic prediction.
- **STAEFormer** [35]: STAEformer uses spatio-temporal adaptive embedding to capture the intrinsic spatio-temporal dependencies.

- **PGCN** [42]: PGCN builds a progressive adjacency graph and employs dilated causal convolution to extract spatial-temporal features.
- **STPGNN** [16]: STPGNN enhances prediction performance by identifying pivotal nodes in the road network. The pivotal graph convolution module is design to capture complex spatial-temporal dependencies.
- **HimNet** [43]: HimNet captures spatiotemporal heterogeneity through spatial and temporal embeddings and learns spatiotemporal-specific parameters from meta-parameter pools.
- **STAMT** [44]: STAMT leverages a memory bank to cache and query historical traffic patterns for better spatial-temporal correlation extraction while reducing computational costs through efficient temporal aggregation modules.

D. Experimental Setups

The model is trained in a Pytorch environment with one NVIDIA V100 with 32GB memory (GPU). In regular traffic forecasting task, we set the input and output length to 12 while in long-term task the length is set to 60. The initial learning rate to 0.001 and the batch size is set to 64. We use Ranger21 optimizer [59] in the training process. The model employs 2 temporal blocks and local kernel size sets to 3. The feature dimension C is set to 128. For the spatial component, we use a single spatial block, with the hyperparameter e set to 5. In addition, we randomly shuffle the training data at the beginning of each epoch to ensure that the model learns the underlying patterns.

E. Performance comparison (RQ1)

We design and conduct three types of experiments to validate the model's performance, including regular, long-term and large-scale traffic forecasting tasks. Regular traffic forecasting typically involves experiments on widely used standard datasets to evaluate the fundamental performance of models, which is the primary focus of most current research. This type of forecasting is crucial for real-time applications, enabling timely responses to dynamic traffic conditions. In contrast, long-term traffic forecasting assesses a model's capabilities over extended periods, such as predicting 60-time steps. This aspect is important because long-term forecasts are essential for strategic planning, such as route optimization, infrastructure development, and traffic management in urban areas. The large-scale forecasting task evaluates the model's scalability and computational efficiency, ensuring its applicability to real-world scenarios characterized by high spatial and temporal complexity.

The results and analysis are as follows:

1) *Results and analysis on regular traffic forecasting:* Table IV and Table V present the performance comparison on six traffic datasets. The baseline model covers statistical methods, STGNNs, and attention-based methods.

On highway datasets, all deep learning models are superior to the statistical methods, indicating the effectiveness of deep learning methods in spatial-temporal modeling. Early

TABLE IV
PERFORMANCE COMPARISON ON HIGHWAY DATASETS. **BOLD**: BEST, UNDERLINE: SECOND BEST.

Method	PEMS03			PEMS04			PEMS07			PEMS08		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
HA	31.58	52.39	33.78%	38.03	29.24	27.88%	45.12	65.64	24.51%	34.86	59.24	27.88%
VAR	23.65	38.26	24.51%	28.70	44.56	19.20%	50.22	75.63	32.22%	19.19	29.81	13.10%
STGCN	17.55	30.42	17.34%	21.16	34.89	13.83%	25.33	39.34	11.21%	17.50	27.09	11.29%
DCRNN	18.39	30.55	20.24%	21.22	33.44	14.17%	25.22	38.61	11.82%	16.82	26.35	10.92%
GraphWaveNet	19.12	32.75	18.86%	24.89	39.66	17.29%	26.39	41.52	11.96%	18.28	30.05	12.15%
ASTGCN	18.29	30.80	17.05%	22.93	35.22	16.56%	25.98	39.65	11.84%	18.86	28.74	11.15%
AGCRN	15.20	26.57	14.64%	19.95	32.91	13.16%	20.57	34.40	8.74%	15.81	25.04	10.19%
STG-NCDE	15.57	27.09	15.06%	19.21	31.09	12.76%	20.53	33.84	8.80%	15.45	24.81	9.91%
ASTGNN	14.81	24.91	14.83%	18.74	30.95	12.47%	19.26	32.75	8.54%	15.32	24.61	9.64%
Ada-STNet	14.49	25.11	<u>14.35%</u>	18.65	29.95	13.09%	21.44	34.26	9.92%	14.21	23.02	9.29%
DGCRN	14.51	24.81	15.11%	19.22	30.58	13.44%	20.96	33.35	9.55%	14.97	23.72	9.78%
STAEFormer	15.18	25.80	15.33%	18.21	30.11	12.00%	19.15	32.61	8.02%	13.55	23.01	8.98%
PGCN	14.94	25.54	15.06%	18.68	30.08	13.07%	20.77	33.81	8.99%	14.22	23.00	9.30%
STPGNN	<u>14.41</u>	24.48	15.10%	18.57	29.89	12.97%	20.00	32.82	8.49%	14.17	23.25	9.21%
HimNet	15.42	27.35	15.59%	18.20	30.07	<u>12.06%</u>	19.25	32.79	8.08%	13.86	23.23	9.25%
STAMT	14.61	23.96	15.33%	<u>18.07</u>	<u>29.53</u>	12.11%	<u>19.08</u>	<u>32.13</u>	<u>8.05%</u>	13.40	<u>22.70</u>	8.86%
TWIST	14.40	<u>24.01</u>	14.31%	18.04	29.52	12.07%	18.97	32.01	8.06%	<u>13.47</u>	22.68	<u>8.89%</u>

TABLE V
PERFORMANCE COMPARISON ON TRAFFIC DEMAND DATASETS. **BOLD**: BEST, UNDERLINE: SECOND BEST.

Method	NYCTaxi pick			NYCTaxi drop			NYCBike pick			NYCBike drop		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
STGCN	6.11	11.04	41.03%	5.76	11.95	39.75%	2.17	3.62	55.76%	2.10	3.31	51.19%
DCRNN	5.88	10.71	37.09%	5.19	9.63	37.78%	2.09	3.30	54.22%	1.96	2.94	51.42%
GraphWaveNet	5.88	10.09	42.67%	5.25	9.29	36.46%	2.18	3.47	56.25%	2.09	3.26	53.78%
ASTGCN	6.78	14.92	47.88%	6.55	12.75	42.80%	2.14	3.97	61.73%	2.04	3.60	60.09%
AGCRN	6.21	12.55	41.50%	5.91	14.00	40.89%	2.07	3.20	51.92%	2.14	3.46	56.56%
STG-NCDE	6.11	11.35	47.37%	5.99	10.62	43.08%	2.28	4.15	62.59%	2.18	3.85	61.66%
ASTGNN	5.91	10.71	40.17%	6.28	12.05	49.78%	2.27	3.34	57.23%	2.36	3.67	60.09%
Ada-STNet	5.55	9.70	36.95%	5.45	9.73	36.55%	2.22	3.60	53.71%	2.05	3.18	50.73%
DGCRN	5.63	9.96	39.34%	5.45	9.73	36.43%	2.52	3.99	58.76%	2.47	3.65	60.86%
STAEFormer	5.20	9.17	35.99%	4.95	8.81	33.98%	2.17	3.42	56.21%	2.14	3.30	54.75%
PGCN	6.49	11.54	42.95%	5.68	9.97	40.28%	2.29	3.72	56.46%	2.19	3.47	54.44%
STPGNN	7.04	13.16	53.40%	7.31	13.46	52.54%	2.28	3.70	57.35%	2.16	3.38	54.88%
HimNet	5.78	10.56	37.12%	5.82	11.63	39.38%	2.32	3.69	63.21%	2.18	3.40	58.99%
STAMT	4.98	8.74	33.26%	4.79	8.43	34.29%	<u>1.99</u>	<u>3.10</u>	52.87%	<u>1.88</u>	<u>2.81</u>	50.43%
TWIST	<u>5.18</u>	<u>9.04</u>	<u>35.70%</u>	<u>4.93</u>	<u>8.75</u>	33.64%	1.97	3.08	<u>52.07%</u>	1.87	2.79	51.31%

spatial-temporal graph neural networks, such as STGCN and DCRNN, rely on predefined adjacency matrices, which hinder their ability to capture global spatial information, leading to poor performance. AGCRN outperforms GraphWaveNet and ASTGCN due to the improved adaptive graph learning algorithm. ASTGNN is a transformer-based method and outperforms the classic STGNNs. Ada-STNet constructs graph structures from both macro and micro perspectives, achieving superior performance compared to previous methods. STAEFormer leverages novel adaptive embedding to enhance the performance of the vanilla Transformer, achieving competitive results across all highway datasets except PEMS03. PGCN constructs progressive adjacency matrices that dynamically adapt the graph structure based on input data, ensuring robust performance across diverse datasets. HimNet and STAMT also demonstrate superior performance on highway datasets.

On traffic demand datasets, we only compared deep learning methods. From the result, we can observe that STGCN exhibits competitiveness on NYC-Bike datasets, indicating its effectiveness. The same phenomenon applies to DCRNN and GrapWaveNet. We find that while some models exhibit slightly lower performance on highway datasets (coarse-

grained), they perform notably better on traffic demand datasets (fine-grained). This discrepancy can be attributed partly to the inherent characteristics of the datasets and partly to the model architectures. For example, earlier models with simpler designs tend to be better adapted to fine-grained datasets with smaller data volumes. Similar to highway datasets, Ada-STNet, DGCRN and STAEFormer also demonstrate strong capabilities in modeling traffic dynamics, showcasing their adaptability across various scenarios. PGCN and STPGNN perform well on the NYC-Bike dataset but show slightly weaker performance on the NYC-Taxi dataset. HimNet performs well on highway data, but ineffectively on traffic demand datasets. Compared to HimNet, STAMT demonstrates superior performance, proving its effectiveness and generalization.

From these tables, we have following observations: (1) Our TWIST achieves the best or near-best performance on regular traffic forecasting tasks. This success can be attributed to its temporal trend-aware window attention, which effectively captures the temporal dynamics in traffic series. Additionally, the sparse attention mechanism facilitates global spatial feature extraction, enabling accurate predictions. (2) The traditional

TABLE VI
PERFORMANCE COMPARISON ON LONG-TERM TRAFFIC FORECASTING. **BOLD:** BEST, UNDERLINE: SECOND BEST.

Dataset	Method	@Horizon 36			@Horizon 48			@Horizon 60			Average		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PEMS04	STGCN	23.90	37.28	16.98%	24.70	38.53	17.85%	26.43	40.80	19.45%	23.16	36.11	16.93%
	DCRNN	23.67	37.62	16.03%	24.47	38.86	16.81%	25.39	39.92	18.10%	22.79	36.19	15.70%
	ASTGCN	31.01	46.77	20.91%	33.15	49.82	23.16%	38.53	57.06	28.91%	29.97	45.76	20.54%
	AGCRN	23.33	38.34	15.48%	23.90	39.18	16.02%	25.72	41.42	17.66%	22.76	37.23	15.21%
	STG-NCDE	24.27	38.50	16.65%	24.81	39.26	17.32%	26.24	41.25	19.18%	23.56	37.23	16.45%
	ASTGNN	24.12	40.20	15.45%	24.90	41.64	16.17%	25.96	43.21	17.12%	23.00	38.34	14.90%
	DGCRN	23.74	37.36	16.64%	28.62	42.85	21.60%	38.68	57.36	31.15%	24.59	38.90	17.68%
	STAEFormer	23.31	37.80	14.65%	23.89	38.71	15.02%	24.47	39.47	<u>15.58%</u>	22.41	36.36	<u>14.29%</u>
	PGCN	25.71	40.18	17.48%	26.51	41.10	18.28%	28.38	43.76	20.63%	24.51	38.12	17.03%
	STPGNN	25.05	38.64	17.80%	25.92	39.84	18.21%	27.58	41.99	19.78%	24.02	37.22	17.05%
	HimNet	20.92	35.42	13.55%	21.45	36.41	13.85%	22.01	37.29	14.27%	<u>20.35</u>	34.32	13.30%
	STAMT	21.42	34.96	15.79%	21.89	35.67	16.05%	22.78	36.91	16.83%	<u>20.78</u>	33.76	15.33%
	TWIST	20.85	34.15	<u>14.57%</u>	21.38	34.92	<u>14.80%</u>	<u>22.27</u>	36.02	<u>15.61%</u>	20.30	33.12	14.41%
PEMS08	STGCN	21.32	33.03	14.70%	23.05	35.35	15.69%	24.88	37.88	17.19%	20.47	31.64	14.05%
	DCRNN	20.48	31.73	13.73%	21.38	33.03	14.56%	23.12	35.22	16.35%	20.02	30.93	13.58%
	ASTGCN	29.32	42.91	19.76%	31.63	46.27	21.19%	36.10	52.47	25.35%	27.21	40.87	18.55%
	AGCRN	21.29	33.84	14.50%	22.53	35.78	15.52%	24.73	38.79	17.15%	20.57	32.75	13.93%
	STG-NCDE	22.20	34.28	15.99%	22.73	35.32	15.93%	23.57	36.52	16.67%	21.00	32.58	14.99%
	ASTGNN	21.23	33.38	14.82%	22.22	35.09	15.87%	23.55	36.56	17.06%	20.58	32.29	14.54%
	DGCRN	18.84	30.99	12.65%	23.68	36.47	16.83%	33.45	49.61	26.47%	19.67	32.35	13.62%
	STAEFormer	17.62	29.39	13.09%	17.89	29.93	12.33%	18.82	30.76	13.83%	16.95	28.26	12.01%
	PGCN	19.82	31.99	13.42%	20.86	33.60	14.17%	22.30	35.59	15.66%	19.20	30.74	13.42%
	STPGNN	19.56	31.79	13.03%	20.46	33.29	13.93%	21.92	35.21	15.17%	18.98	30.70	12.80%
	HimNet	17.97	31.29	11.88%	18.48	32.41	<u>12.25%</u>	19.05	33.41	12.66%	17.24	29.83	11.38%
	STAMT	<u>17.31</u>	<u>28.60</u>	<u>12.12%</u>	17.94	<u>29.52</u>	12.78%	<u>18.60</u>	<u>30.43</u>	13.49%	<u>16.67</u>	<u>27.36</u>	11.74%
	TWIST	16.99	28.58	11.58%	17.57	29.41	12.14%	18.16	30.18	<u>12.77%</u>	16.28	27.17	11.11%

methods HA and VAR perform worse among these datasets, the high errors show their inability to address complex traffic forecasting tasks. (3) Models that consider dynamic spatial structures (e.g., ASTGNN, Ada-SNTet, and TWIST) tend to outperform models with static structures (e.g., STGCN, DCRNN, and GraphWaveNet) because they can more accurately capture the dynamic spatial correlations. (4) Although some methods perform well on highway datasets (e.g., PGCN, STPGNN), their performance is decreased on traffic demand datasets. This indicates that while these methods can handle coarse-grained traffic scenarios effectively, they struggle to deal with fine-grained urban traffic demand data. We argue that the TAWA module in TWIST is key to its ability to maintain competitive performance across both coarse-grained and fine-grained traffic forecasting scenarios. This mechanism dynamically captures temporal patterns at different time scales, allowing the model to adapt to various temporal characteristics.

2) *Results and analysis on long-term traffic forecasting:* As mentioned above, the trend-aware window attention module in TWIST efficiently captures temporal dependencies in traffic data, enhancing prediction performance while maintaining scalability and computational efficiency, even for extended time horizons. Table VI presents the results of long-term forecasting experiments, comparing TWIST with baseline methods on the PEMS04 and PEMS08 datasets over a 5-hour horizon.

The performance trends in long-term forecasting are consistent with those in short-term forecasting. However, as shown in the table, all methods exhibit a significant performance decline when transitioning from short-term to long-term predictions, primarily due to increased uncertainty introduced by larger historical time windows, which amplifies prediction errors.

Table VII compares the error changes of STAEFormer, HimNet, STAMT, and TWIST during the transition from

TABLE VII
COMPARISON OF ERRORS CHANGE FROM SHORT-TERM TRANSITIONING TO LONG-TERM PREDICTIONS. **BOLD:** BEST, UNDERLINE: SECOND BEST.

Dataset	Method	Metrics		
		MAE	RMSE	MAPE
PEMS04	STAEFormer	23.06%	20.78%	<u>19.08%</u>
	HimNet	11.81%	14.13%	10.10%
	STAMT	14.99%	14.32%	26.58%
	TWIST	<u>12.53%</u>	11.83%	19.41%
PEMS08	STAEFormer	25.10%	22.83%	33.74%
	HimNet	24.39%	28.42%	23.03%
	STAMT	24.40%	20.52%	32.50%
	TWIST	20.88%	19.78%	<u>24.99%</u>

short-term to long-term predictions. TWIST and HimNet exhibit minimal error growth, demonstrating their stability and generalization capabilities in both regular and long-term forecasting tasks. In contrast, while STAMT performs well in short-term predictions, it shows limitations in long-term forecasting, likely due to its temporal aggregation convolution (TAC), which may lose critical information essential for capturing long-range dependencies. Notably, different models exhibit varying degrees of error growth, which can be partially attributed to dataset-specific characteristics, as each dataset presents unique data distributions. In addition, the difference in feature extraction capability of each model also contributes to this discrepancy. While certain models and architectures excel in short-term forecasting, they often struggle to capture long-term dependencies. To mitigate biases arising from diverse data distributions and model architectures, our future work will focus on developing adaptive mechanisms that generalize effectively across datasets while ensuring the model's robustness to varying traffic patterns.

In Fig. 4, we visualize the prediction error metrics for each time interval on six datasets. The results show that our model exhibits relatively stable changes, further indicating its

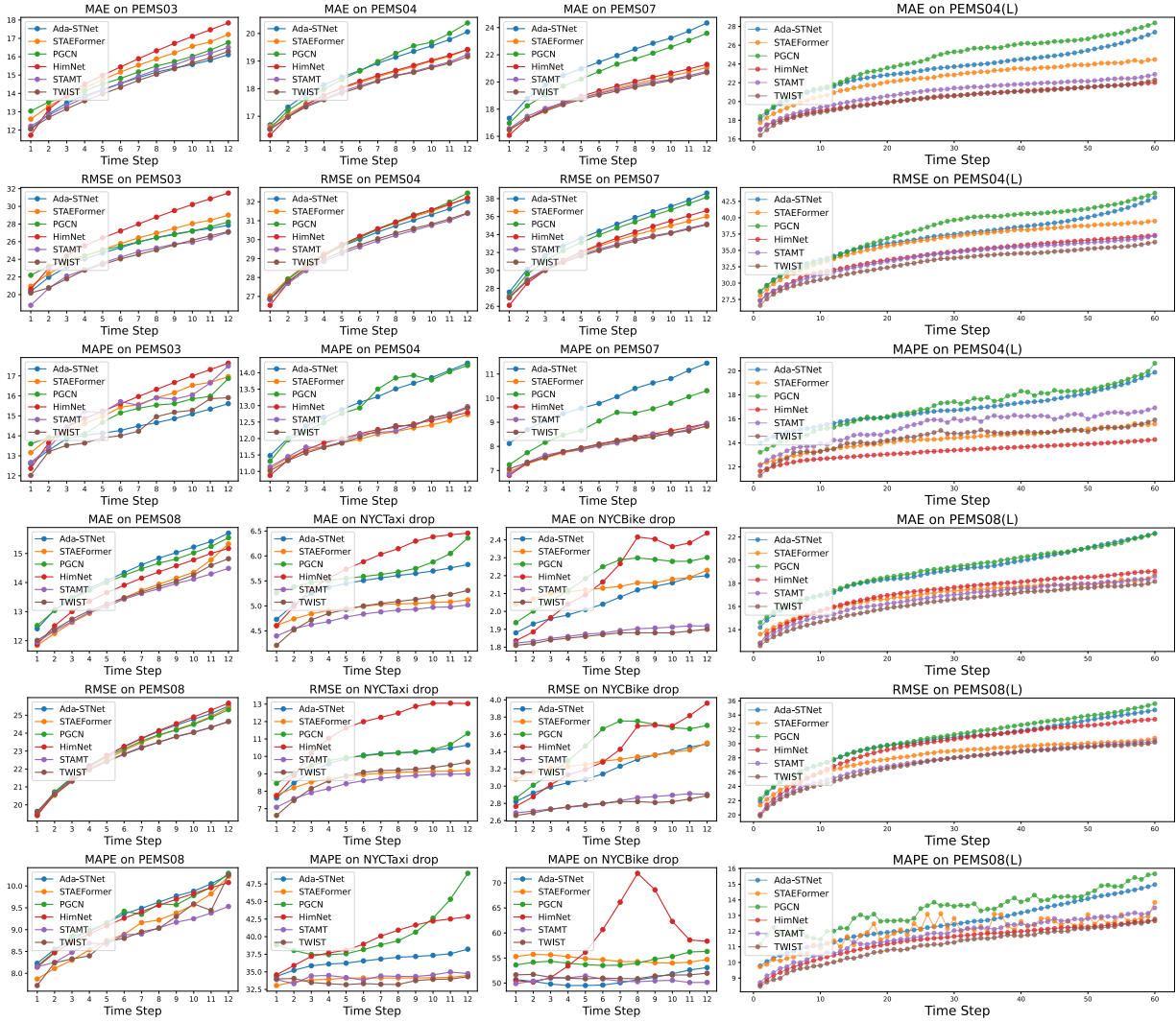


Fig. 4. Performance changes of different methods on the six datasets as the prediction interval increases.

TABLE VIII
PERFORMANCE COMPARISON ON LARGE-SCALE TRAFFIC FORECASTING. **BOLD: BEST, UNDERLINE: SECOND BEST.**

Dataset	Method	@Horizon 3			@Horizon 6			@Horizon 12			Average		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Carpark	STGCN	4.48	20.82	12.00%	5.43	21.75	13.79%	8.44	24.93	17.40%	5.92	22.36	13.98%
	DCRNN	2.61	7.73	7.01%	4.21	10.11	10.93%	6.33	14.24	14.08%	4.24	10.57	10.60%
	GraphWaveNet	2.72	8.02	7.77%	4.27	10.40	10.95%	6.45	14.51	14.52%	4.25	10.64	10.58%
	ASTGCN	3.55	13.30	7.89%	5.19	15.51	11.63%	8.04	19.99	15.53%	5.33	16.11	11.22%
	AGCRN	4.76	14.04	6.57%	6.58	15.09	11.16%	8.06	15.37	13.66%	6.58	17.33	10.80%
	STG-NCDE	3.83	10.21	9.61%	5.24	12.69	12.88%	8.45	18.21	17.09%	5.59	13.68	12.66%
	Ada-STNet	2.88	9.41	8.91%	4.34	11.63	13.52%	6.27	15.22	16.65%	4.33	11.77	12.55%
	DGCRN	2.33	<u>7.58</u>	6.70%	<u>3.95</u>	<u>9.91</u>	<u>10.87%</u>	<u>5.66</u>	13.38	14.12%	<u>3.80</u>	<u>9.98</u>	10.07%
	STAEFormer	2.78	8.37	9.13%	4.14	10.16	12.40%	5.69	<u>13.31</u>	15.06%	4.03	10.54	11.81%
	PGCN	2.56	8.06	7.27%	4.26	10.52	11.38%	6.55	14.74	15.00%	4.23	10.74	10.86%
	STPGNN	2.77	7.85	6.83%	4.24	10.34	11.10%	6.39	14.50	14.54%	4.22	10.51	10.31%
	STAMT	<u>2.30</u>	7.95	<u>6.33%</u>	4.15	12.94	11.01%	6.36	29.47	14.14%	4.03	15.86	<u>9.95%</u>
	TWIST	2.25	7.52	6.28%	3.87	9.41	10.80%	5.62	12.59	14.16%	3.70	9.49	9.90%
GBA	STGCN	21.05	34.51	16.42%	23.63	38.92	18.35%	26.87	44.45	21.92%	23.42	38.57	18.46%
	DCRNN	18.71	30.36	14.72%	23.06	36.16	20.45%	29.85	46.06	29.93%	23.13	36.35	20.84%
	GraphWaveNet	17.85	29.12	13.92%	21.11	33.69	17.79%	25.58	40.19	23.48%	20.91	33.41	17.66%
	ASTGCN	21.46	33.86	17.24%	26.96	41.38	24.22%	34.29	52.44	32.53%	26.47	40.99	23.65%
	AGCRN	18.31	30.24	14.27%	21.27	34.72	16.89%	24.85	40.18	20.80%	21.01	34.25	16.90%
	DGCRN	18.02	29.49	14.13%	21.08	34.03	16.94%	25.25	40.63	21.15%	20.91	33.83	16.88%
	STAMT	<u>16.74</u>	<u>28.19</u>	<u>13.09%</u>	<u>19.53</u>	<u>32.39</u>	<u>15.67%</u>	<u>23.43</u>	<u>38.23</u>	<u>19.55%</u>	<u>19.41</u>	<u>32.18</u>	<u>15.71%</u>
	TWIST	16.56	28.02	12.11%	19.37	32.36	14.93%	23.11	37.83	18.86%	19.19	31.87	15.43%

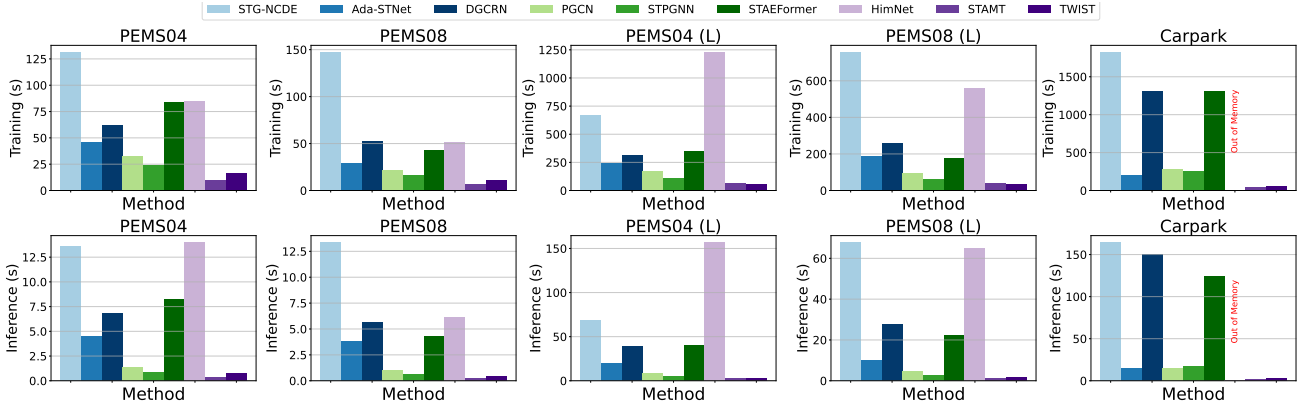


Fig. 5. Computation efficiency comparison on three traffic datasets. PEMS04 (L) and PEMS08 (L) denote the long-term traffic forecasting tasks on PEMS04 and PEMS08. HimNet is excluded from Carpark due to the OOM issue.

reliability and generalization in standard and long-term traffic forecasting tasks.

3) Results and analysis on large-scale traffic forecasting:

Table VIII presents the performance comparison of various models on large-scale traffic datasets. These datasets, characterized by large node counts and dense spatial correlations, present significant challenges for accurate predictions. Notably, constrained by limited computational resources, some baseline models were excluded from comparisons due to out-of-memory (OOM) issues during the training process. Experimental results demonstrate that TWIST consistently achieves state-of-the-art performance on both the Carpark and GBA datasets, highlighting its ability to effectively model complex spatiotemporal dependencies in large-scale scenarios, overcoming the challenges posed by high node counts and dense spatial correlations. This is attributed to its efficient spatial attention mechanism, which balances computational efficiency with model expressiveness, and its trend-aware temporal attention mechanism, which enhances adaptability to diverse and dynamic traffic patterns in large-scale datasets. Interestingly, while models like DGCRN show competitive performance on the Carpark dataset, their effectiveness diminishes on the more complex GBA dataset, reflecting their limited adaptability to intricate spatial correlations. Similarly, methods such as STGCN and DCRNN, which rely on predefined spatial structures, struggle to capture global dependencies, limiting their effectiveness in representing the dynamic and complex spatial correlations inherent in large-scale traffic networks.

F. Efficiency study (RQ2)

As research advances, prediction models have grown increasingly complex, enabling them to process larger datasets and achieve superior performance across diverse traffic scenarios. However, this complexity has raised significant concerns regarding computational efficiency, particularly in long-term and large-scale forecasting tasks. Therefore, striking a balance between accuracy and efficiency is critical. In Fig. 5, we compare the computational efficiency of TWIST with eight state-of-the-art methods across three datasets, encompassing regular, long-term, and large-scale forecasting tasks.

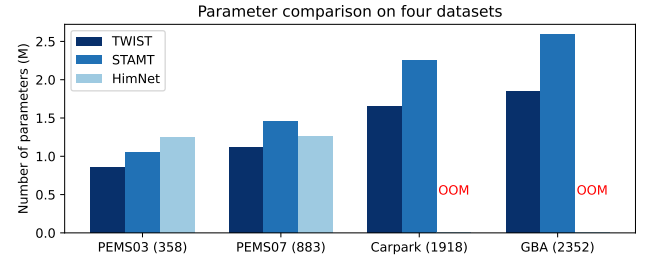


Fig. 6. Number of parameters comparison on four datasets as the nodes increase.

The results reveal that STG-NCDE and HimNet exhibit prolonged training and inference times due to their high complexity and substantial computational demands, particularly when processing large datasets. While HimNet achieves competitive performance on different forecasting tasks, it fails to balance accuracy and efficiency, limiting its practicality in real-world scenarios where both are essential. Ada-STNet and STPGNN demonstrate moderate computational performance, whereas PGCN achieves high efficiency at the expense of accuracy. STAEFormer achieves competitive performance in long-term forecasting but at the cost of significantly higher computational demands. In terms of short-term forecasting, both STAMT and TWIST exhibit competitive performance and efficiency. However, TWIST outperforms other methods in long-term forecasting by achieving an optimal balance between performance and computational efficiency. To further evaluate the model's consumption of memory resources, we visualized the number of parameters of the model on four datasets, as shown in Fig. 6. It is observed that our TWIST demonstrates a significantly lower number of parameters. This can be attributed to our simple architecture and optimized attention mechanism. In contrast, the reason for the higher parameter count in STAMT can be attributed to its memory bank mechanism, which stores historical traffic patterns and relies on extensive parameterized components for pattern management and retrievals. For HimNet, it employs meta-learning techniques, which require additional parameters for task adaptation, resulting in excessive memory consumption.

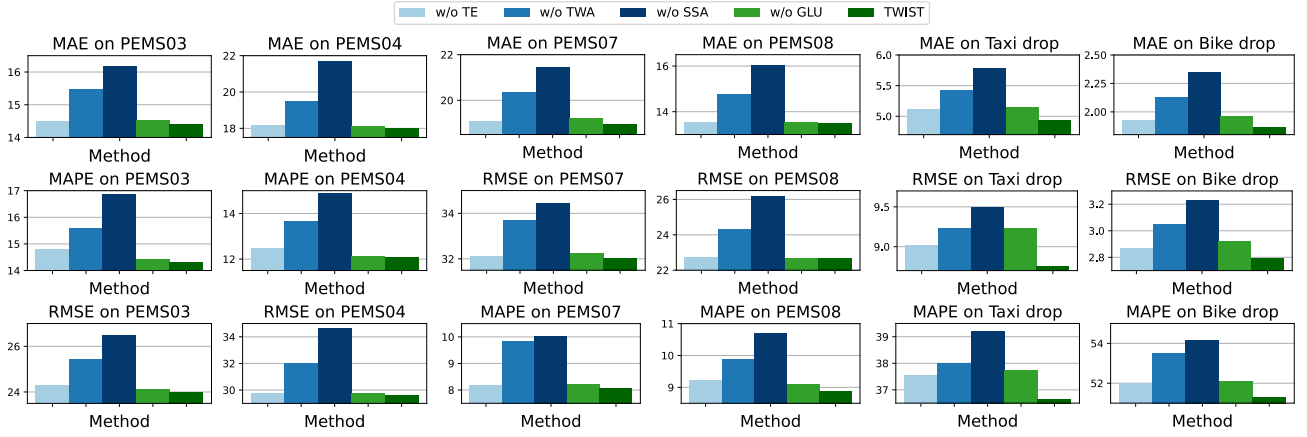


Fig. 7. Ablation study on six traffic datasets.

that limits its efficiency and suitability for large-scale traffic prediction tasks.

G. Ablation study (RQ3)

To demonstrate the validity of the different modules in TWIST, we compare TWIST with the following variants:

- **w/o TE:** remove the temporal embedding layer.
- **w/o TAWA:** remove trend-aware window attention.
- **w/o SSA:** remove sparse spatial attention and no longer considers spatial dependencies.
- **w/o GLU:** remove the gated linear unit.

We conduct ablation studies on six real-world traffic datasets, as shown in Fig. 7. From the results, we can draw the following insights: (1) Removing TE increases errors across all datasets, highlighting the significance of temporal embedding to accentuate periodicity's influence. Without TE, TWIST cannot fully exploit the time-related regularities, leading to degraded performance, particularly in scenarios where periodic patterns dominate traffic behavior. (2) Without TAWA, the model's performance drops significantly, indicating that TAWA is effective in capturing temporal dependencies. The TAWA module enhances the model's ability to focus on relevant time windows and capture local and global temporal dynamics. Its removal results in a noticeable performance decline, suggesting that TWIST loses its capacity to leverage multi-scale temporal dependencies. (3) Removing SQA leads to the declined performance, showing the significance for spatial feature extraction. Its absence results in a weakened ability to model spatial interactions, leading to performance degradation across datasets. This decline emphasizes the importance of capturing spatial dependencies in traffic networks, as traffic flow at different locations is highly interdependent. (4) Removing GLU slightly degrades the performance, suggesting that it fine-tunes the output by selectively passing on key information.

H. Comparative Study

In Fig. 8, to provide a more detailed analysis of the effectiveness and superiority of the proposed TAWA and SQA

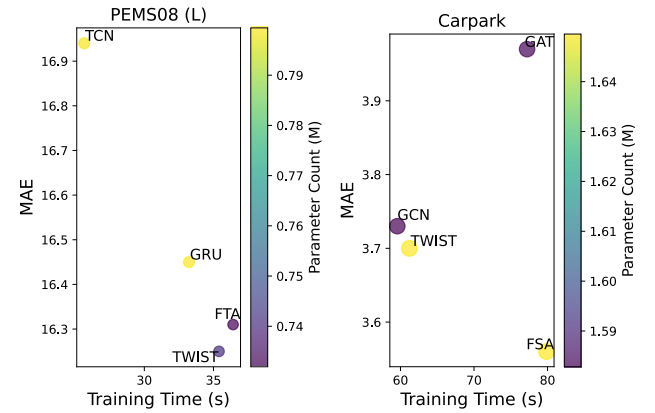


Fig. 8. Comparison of different variants on long-range forecasting tasks.

compared to existing methods, we conduct a comparative study by systematically replacing these components with commonly used approaches, including GRU [8], TCN [16], full temporal attention (FTA, removing the trend-aware and window mechanism) in temporal dimension, and GCN [16], GAT [60], full spatial attention (FSA, removing the sparse mechanism) in spatial dimension. To ensure a fair comparison, all variants adopt the same hyperparameter settings, including the number of spatial-temporal layers, channel dimensions, and attention heads. To eliminate biases introduced by predefined graph structures, we derive the adjacency matrix from learnable vectors [9]. This study aims to assess whether TAWA and SQA offer meaningful improvements compared to widely used methods in long-term and large-scale traffic forecasting tasks. The variants are as follows:

- **Replacing TAWA with GRU:** Replace TAWA with the gated recurrent unit to capture long-term dependencies, but it may encounter gradient vanishing or explosion.
- **Replacing TAWA with TCN:** Replace TAWA with TCN, which uses dilated causal convolutions to capture local temporal relationships.
- **Replacing TAWA with FTA:** Replace TAWA with full temporal self-attention to allow the model to attend to

all time steps and evaluate its effect on efficiency and performance.

- Replacing SQA with GCN: Replace SQA with GCN, where node representations are updated using localized graph convolutions based on neighbors.
- Replacing SQA with GAT: Replace SQA with GAT, which uses learnable attention coefficients for dynamic spatial aggregation.
- Replacing SQA with FSA: Replace SQA with full spatial self-attention to allow global attention across all nodes, at the cost of higher computational complexity.

As shown in Fig. 8, TWIST balances prediction performance and efficiency. Specifically, using GRU or TCN instead of TAWA increases errors, likely GRU tends to suffer from gradient vanishing or explosion issues in long sequences, limiting its long-range memory capability. Although the TCN-based approach is slightly more computationally efficient in our experiment design, the two-layer TCN struggles to capture enough temporal features, resulting in poor performance. FTA employs full temporal attention, it requires higher computational costs without better results. For spatial modeling, although graph convolution-based methods are more efficient, their computational errors are large, which may be due to the learned static adjacency matrix. Moreover, GCNs usually need to stack multiple layers to capture long-distance dependencies. GAT also needs longer training time due to the point-wise dot product operations. FSA uses complete spatial attention to model global dependencies, but it greatly increases computation time. These results demonstrate that SQA's sparse design optimally balances efficiency and accuracy. Overall, TWIST shows the best performance and efficiency for long-range traffic prediction, proving our methods' effectiveness.

I. Rubost testing (RQ5)

Data collection is often impacted by various factors such as sensor malfunctions, communication errors, or environmental disturbances, all of which can lead to missing data or inaccuracies. Therefore, we expect the model can effectively handle these challenges, ensuring stable and reliable performance. To evaluate the robustness of the model in real-world applications, we design two types of robustness experiments and the results are shown in Fig. 9:

- **Data missing:** To simulate real-world missing data scenarios, we remove different proportions of training data (20%, 40%, and 60%). These ratios correspond to mild, moderate, and severe data loss scenarios, simulating the impact of data loss.
- **Data noise:** To model the uncertainties in real-world data, we introduced varying levels (10%, 20%, and 30%) of random Gaussian noise into the training data. The noise has a mean of 10 and a standard deviation of 500. This approach helps evaluate the model's stability and accuracy when faced with noisy data.

We compare TWIST with DGCRN, STAEFormer, PGCN, and STPGNN on PEMS04 and PEMS08 datasets to evaluate the model's robustness. From Fig. 9, we can observe that TWIST consistently achieves the lowest error (MAE) under

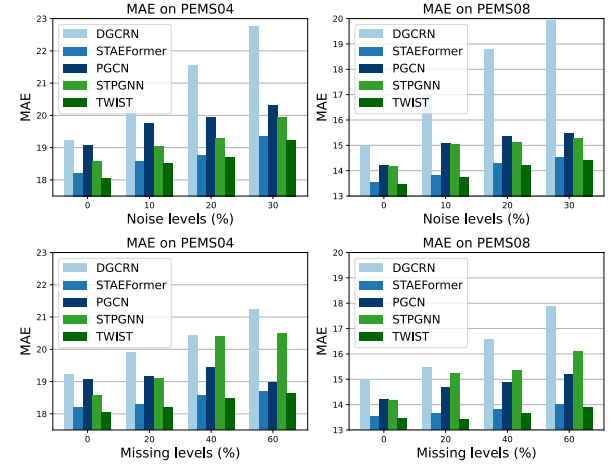


Fig. 9. Robustness testing on PEMS04 and PEMS08 datasets.

all tested conditions. From the results, we can have following findings: (1) DGCRN performs relatively poorly against noise and missing data scenarios, exhibiting the lowest robustness. (2) PGCN and STPGNN show moderate robustness. PGCN shows more robustness in dealing with missing data while STPGNN performs better under noisy conditions. (3) STAEFormer exhibits strong robustness and maintains low errors when dealing with noisy and missing data. (4) TWIST is more robust compared to STAEFormer, indicating it can be widely used in real-world conditions.

The superior robustness of TWIST stems from its architectural design: (1) the trend-aware temporal window mechanism that captures multi-scale trend-level patterns, reducing sensitivity to local fluctuations and ensuring reliable temporal feature extraction even with missing or corrupted data; and (2) the sparse attention with query sampling strategy that distinguishes active and passive queries, enabling the model to focus on broader, reliable patterns while approximating local details, thereby mitigating the impact of noisy or incomplete features.

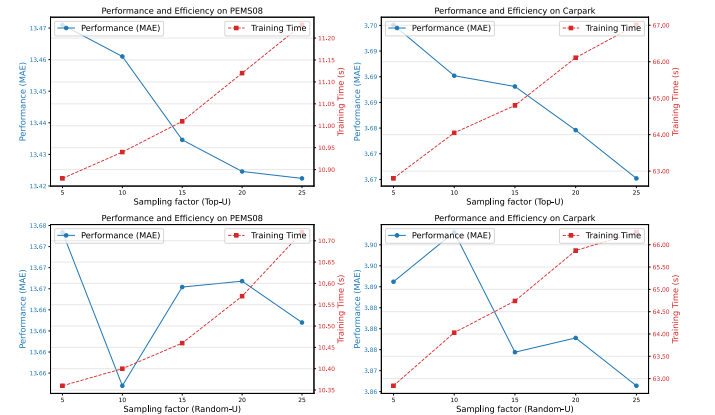


Fig. 10. Parameter analysis experiment on sampling factor e under Top-U and Random-U sampling strategies.

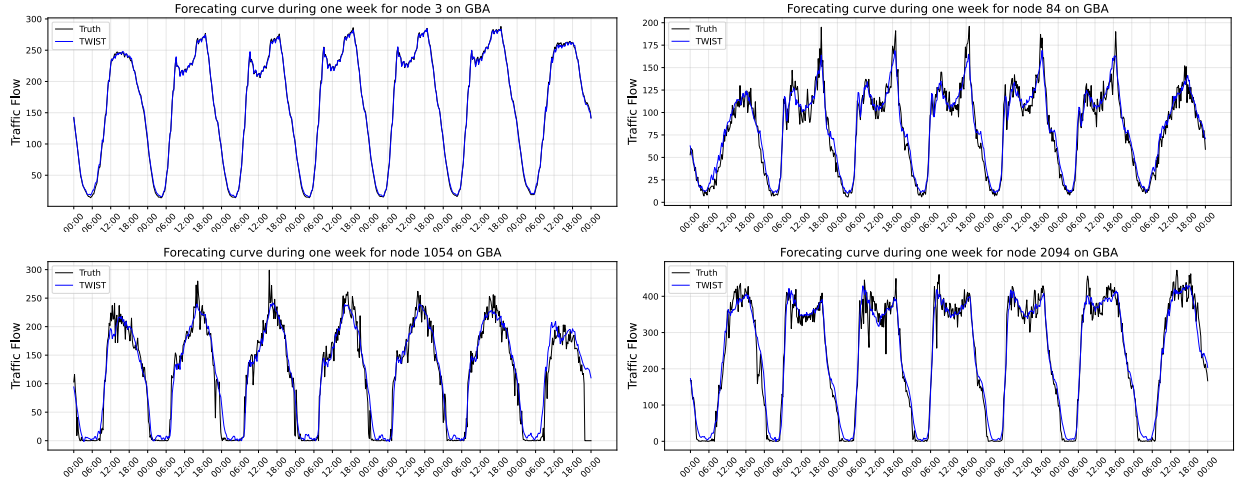


Fig. 11. Visualized comparison between predicted value and ground truth during one week on GBA dataset.

J. Parameter analysis (RQ6)

In Fig. 10, we analyze the impact of the sampling factor on the model's performance. The sampling factor e controls the proportion of queries retained in the spatial attention mechanism, aiming to balance computational efficiency and prediction accuracy. We compared the variations in model performance (MAE) and training time under different sampling factors and strategies on PEMS08 (regular) and Carpark (large-scale) datasets. As observed in the figure, the Top-U sampling strategy consistently outperforms Random-U in terms of prediction accuracy across all sampling factor values. With Top-U, increasing e leads to a steady improvement in performance. This is because Top-U prioritizes nodes with low entropy, capturing the most significant spatial patterns. In contrast, increasing e does not consistently lead to performance improvement with Random-U. This is because random sampling may include redundant or irrelevant node information, which increases the learning noise of the model. In terms of training time, it increases almost linearly with e under these two sampling strategies. Thus using Top-U does not significantly increase additional overhead, especially at lower sampling factors, where its computational efficiency remains sufficient for practical requirements. In short, we can have the following observations: (1) Using Top-U achieves better performance gains, demonstrating the effectiveness of our method. (2) For smaller datasets, the sampling factor e can be appropriately increased to achieve higher prediction accuracy without introducing a significant computational burden. (3) On larger datasets, using Random-U does not reduce resource consumption significantly but instead leads to larger errors, highlighting the importance of selecting a more effective sampling strategy (Top-U) for complex tasks.

K. Case study (RQ7)

In Fig. 11 and Fig. 12, we conduct case studies using the Greater Bay Area (GBA) datasets to explore the performance of our TWIST in real-world scenarios. In Fig. 11, we select multiple nodes from the traffic network and visualize

their ground truth and corresponding predictions generated by TWIST. The results illustrate traffic flow patterns over one week for nodes with diverse characteristics. TWIST accurately predicts the periodic and peak traffic patterns, closely aligning with the ground truth. This demonstrates its ability to model complex spatial-temporal dependencies and adapt to diverse traffic conditions in urban networks.

In Fig. 12, to provide an intuitive comparison of traffic patterns, we visualize the ground truth and predicted values using heatmaps at two timestamps: peak-hour and non-peak-hour traffic flow. Our analysis focuses on three representative regions: Nimitz Freeway (blue mark), Bayshore Freeway (red mark), and San Jose City (purple mark), selected for their diverse traffic patterns and significance within the Greater Bay Area. The heatmaps reveal that TWIST's predictions closely align with the ground truth, particularly in hotspot regions. Notably, peak-hour traffic exhibits higher density and concentration along major freeways and urban centers, such as Nimitz Freeway and San Jose City, reflecting increased congestion. In contrast, non-peak hours exhibit a more dispersed traffic pattern with lower overall intensity, reflecting reduced traffic pressure across the network. These results demonstrate TWIST's ability to accurately capture spatiotemporal dependencies and critical traffic patterns in complex urban scenarios, providing valuable insights for real-world traffic management and planning.

VI. CONCLUSION

To address the challenges of capturing long-range dependencies and modeling complex inter-variable relationships in traffic data, we propose TWIST, an efficient and effective spatial-temporal transformer framework for traffic forecasting. The proposed framework introduces two key innovations: (1) a trend-aware temporal window attention mechanism that captures temporal dependencies over extended periods by leveraging local context to generate queries and keys, while dividing the original sequence into non-overlapping windows to efficiently model local interactions among time steps; and (2) a novel query sampling spatial attention mechanism that

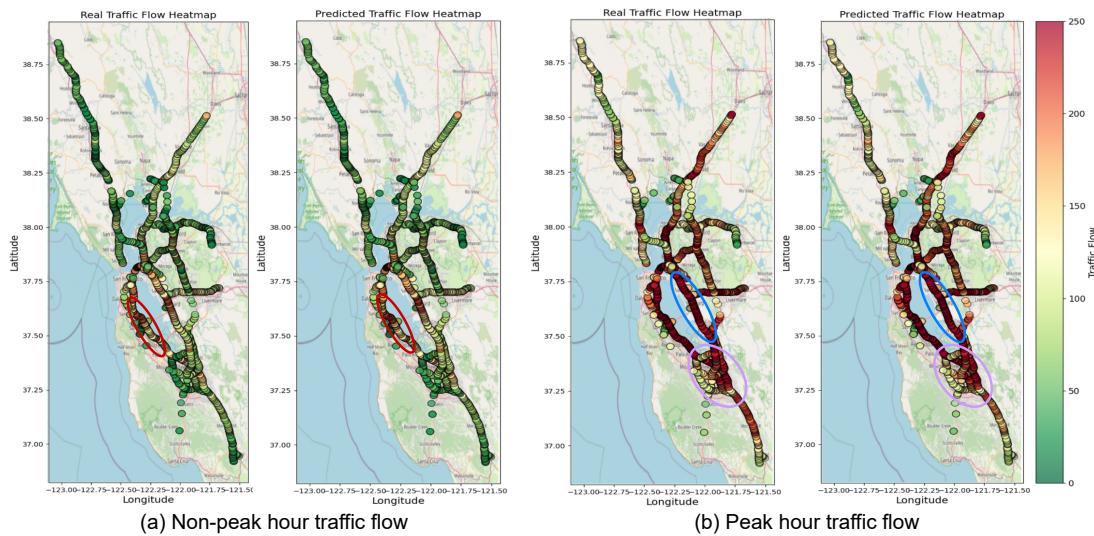


Fig. 12. Traffic flow forecasting visualization on GBA dataset using heatmap. The horizontal and vertical coordinates indicate latitude and longitude, respectively. Red mark: Bayshore Freeway; Blue mark: Nimitz Freeway; Purple mark: San Jose City.

reduces computational complexity while maintaining a global receptive field, enabling effective modeling of spatial correlations. Experiments on eight real-world road datasets show that TWIST consistently outperforms state-of-the-art baselines in regular, long-term and large-scale traffic forecasting tasks.

Our work offers a novel traffic forecasting framework to facilitate research in the field, but it still has some limitations. For example, although the query sampling spatial attention significantly reduces computational complexity from $O(N^2)$ to $O(N \log N)$, further optimization may be required to efficiently scale the model for ultra-large traffic networks (e.g., millions of nodes) or real-time processing scenarios. In addition, the current implementation of TWIST relies primarily on traffic flow or demand data. However, real-world traffic dynamics are influenced by a variety of external factors (e.g., weather, events, and social media signals), which are not integrated into this framework. Future work will explore distributed or federated learning approaches to enhance the framework's scalability for large-scale networks.

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